ProdMatch Final Report

ECE 1786 Creative Applications of Natural Language Processing

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WORD COUNT	
Total	**2061**
Reference	**74**
Word Count	**15**
Total After	**2061-74-15 = 1972**
Penalty	**0%**

Github: https://github.com/ece1786-2024/ProdMatch

1 Introduction

Our project, ProdMatch, is a table recommendation system designed to deliver precise, personalized product recommendations that align with user requirements. By leveraging multimodal models, our system extracts features from product descriptions and images, interprets user queries, and integrates textual and visual data for accurate matching. The project serves two main goals: Precision Matching- analyzing the user's intention and recommending products for the user; Product Analysing - analyzing product information and creating a product database. Machine learning is essential for this task due to its ability to process complex, multimodal data effectively and efficiently.

2 Illustration/ Figure

The system design has two pipelines. The first one uses the VL2-8b model to extract product information and create the product database. The second one uses the same VL2-8b to extract the user query feature path and send it to the matching algorithm to do the product matching.

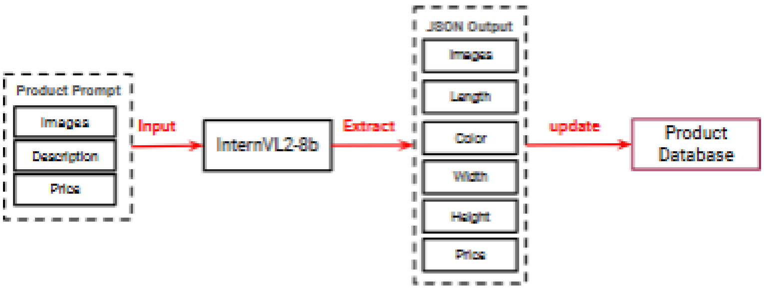


Figure 2.1 Product Database Pipeline

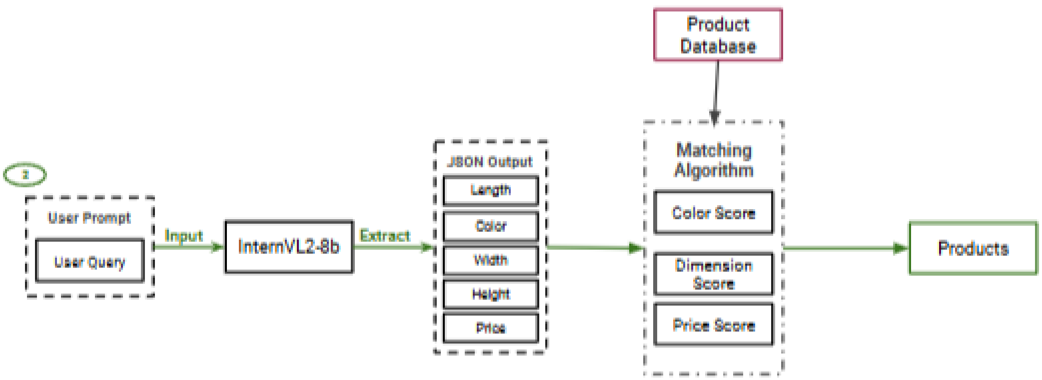


Figure 2.2 Product Matching Pipeline

3 Background & Related Work

The growth of e-commerce has made it challenging for users to find products that match their preferences. Traditional recommendation systems, relying on keywords or collaborative filtering, often fail to integrate visual and textual data effectively. Advancements in multimodal systems and large language models (LLMs) offer solutions by combining diverse data types for precise recommendations.

Lin et al. (2021) introduced the PAM framework, which uses a transformer-based model to process product images, text descriptions, and OCR data, outperforming text-only systems. Similarly, Liu et al. (2023) proposed the Multi-modal Item Embedding Model (MIEM), combining text and image data to improve recommendation relevance and support image-based searches.

Inspired by these works, our project, ProdMatch, leverages fine-tuned multimodal models to align user queries with product attributes. By integrating textual and visual data, ProdMatch addresses existing limitations, advancing the precision and personalization of e-commerce recommendations.

4 Data and Data Processing

In this section, we describe the data collected for the **ProdMatch Project**, the steps taken to clean and format it, and how it was labelled. We created 3 datasets for training and testing in this project.

Product Data: This dataset includes a collection of products **manually** curated from **Amazon** and **Temu** (200 training and 50 tests, manually label the product with features**)**. Each product contains key features, like 1. **Price**: The cost of the product. 2. **Dimensions**: Length, width, and height (potentially expressed in varying units such as inches, centimetres, feet, or meters). 3. **Color**: The product's color.


```
Training Data/      # Folder
├── 01/     # Example
|     ├── .jpg # product image
|     └── .txt # product title
├── ...
├──200/
└── Full.csv # Including the price, Label, and product ID
```

Product Title: Ironalita Solid Wood Dining Table, 47" Mid-Century Modern Natural Wooden, Solid Wood Desk Oak Dinner Table for Dining Room, Kitchen
Price: 129.99
Label: {"colour": "Oak", "length": 47.2, "width": 23.4, "height": 39, "unit": "inch", "price": 129.99}

User Query Data: This dataset consists of 200 training queries and 50 testing queries which are generated by GPT and labeled manually. Each query specifies features like: 1, **Price**: budgets. 2. **Dimensions**: Desired product dimensions (length, width, height). 3. **Color**: Preferred color.

query	label
I'm searching for a black wooden dining table that's about 120 cm long, 75 cm wide, and costs less than $250.	{"color": "Black", "length": "120 cm", "width": "75 cm", "price": 250}
Can I find a dark brown coffee table, around 48 inches long, 30 inches wide, and 18 inches high, under $150?	{"color": "Dark Brown", "length": "48 in", "width": "30 in", "height": "18
Looking for a small work desk in espresso color, roughly 100 cm by 50 cm, for under $100.	{"color": "Espresso", "length": "100 cm", "width": "50 cm", "price": 10
I need a dark wood kitchen table that's 150 cm long, 80 cm wide, and not more than $350.	{"color": "Dark Wood", "length": "150 cm", "width": "80 cm", "price": 3
Can you recommend a dark oak dining table, approximately 6 feet long, with a height of 3 feet, priced below $400?	{"color": "Dark Oak", "length": "6 ft", "height": "3 ft", "price": 400}
I'm looking for a black TV stand, 60 inches wide, 20 inches deep, and no more than $200.	{"color": "Black", "width": "60 in", "depth": "20 in", "price": 200}

Match Test Data: This dataset contains 30 test queries which are designed to for search one specific product, each associated with the correct **product ID**. The product ID corresponds to the most relevant product in the **Product Data** for that query. This dataset is used for evaluating the accuracy of the model's predictions.

Do you have a compact dark gray desk, roughly 55 inches long and 27.5 inches wide, for under $150?	130
Do you offer a desk with a dark coffee finish, dimensions around 63 inches by 31.5 inches, for under $300?	145
A white table with a length of 39.4 inches, width of 18.9 inches, and height of 30.5 inches, priced at $129.99.	34
A maple-colored table, 55 inches long, 28 inches wide, and 28 inches tall, priced at $299.87.	6
An oak table with dimensions: length 51.3 inches, width 19.7 inches, and height 44.7 inches, priced at $99.99.	14

5 Architecture and Software

Our ProdMatch system is a multimodal platform that integrates user input and product data, enabling precise and personalized product recommendations. The model working process can be split into 2 main steps: user prompt processing and product matching.

5.1 User Query Processing

- Input: Users provide natural language descriptions of their needs (e.g., *"I'm looking for a red dining table under $100"*).
- Processing:
    - The query is analyzed using the fine-tuned multimodal model InternVL2-8B, which extracts key features such as:
        - Colour: red
        - Type: dining table
        - Price range: under $100
    - These extracted features are converted into a structured format that matches the format used in the product database, ensuring smooth compatibility for the matching process.

5.2 Product Matching and Recommendation

- Input: Structured features derived from the user's query.
- Processing:
    - Custom matching functions calculate how well each product meets the user's criteria.
    - Multimodal embeddings are employed to align visual and textual features. This ensures products with matching visual and descriptive attributes are prioritized.

- Output: A ranked list of top-matching products is presented to the user.

5.3 Matching Algorithm

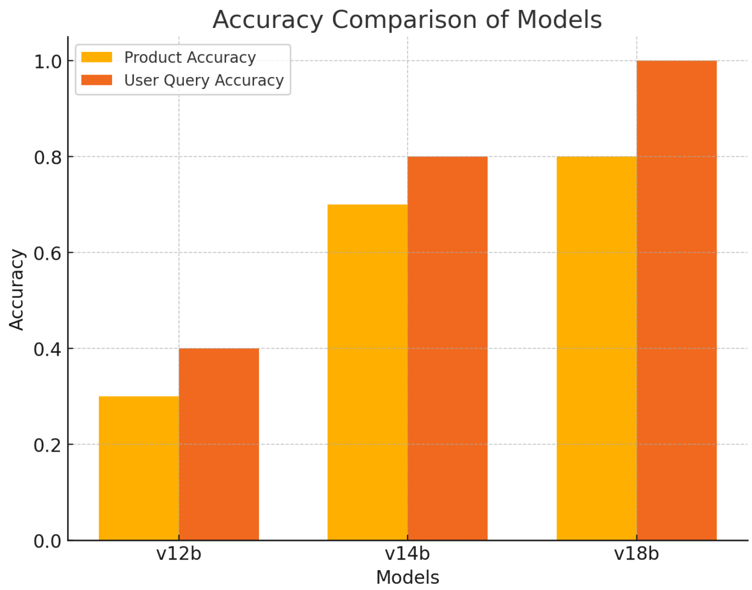
The matching algorithm calculates how relevant each product is to the user's query by combining several scoring metrics into a final score. These metrics include:

- Colour Score:
    - Computed using cosine similarity between text embeddings of product colours.
    - Text embeddings are generated using the pre-trained all-MiniLM-L6-v2 model.
    - This score evaluates the semantic similarity between colour descriptions, accounting for variations in colour terms.
- Dimension Score:
    - Measures the similarity of product dimensions (e.g., length, width, height).
    - Uses a "1 - normalized difference" approach, where smaller differences yield higher scores.
    - Ensures products are physically compatible with the user's requirements.
- Price Score:
    - Evaluate the difference in pricing using the same "1 - normalized difference" method.
    - Ensures that products align with the user's budget.
- Final Score:
    - A weighted sum of the above scores (Color, Dimension, and Price).
    - Each score is assigned a specific weight based on its importance in the matching process.
    - The final score determines the overall quality of the match between the product and the user's query.

6 Baseline Model or Comparison

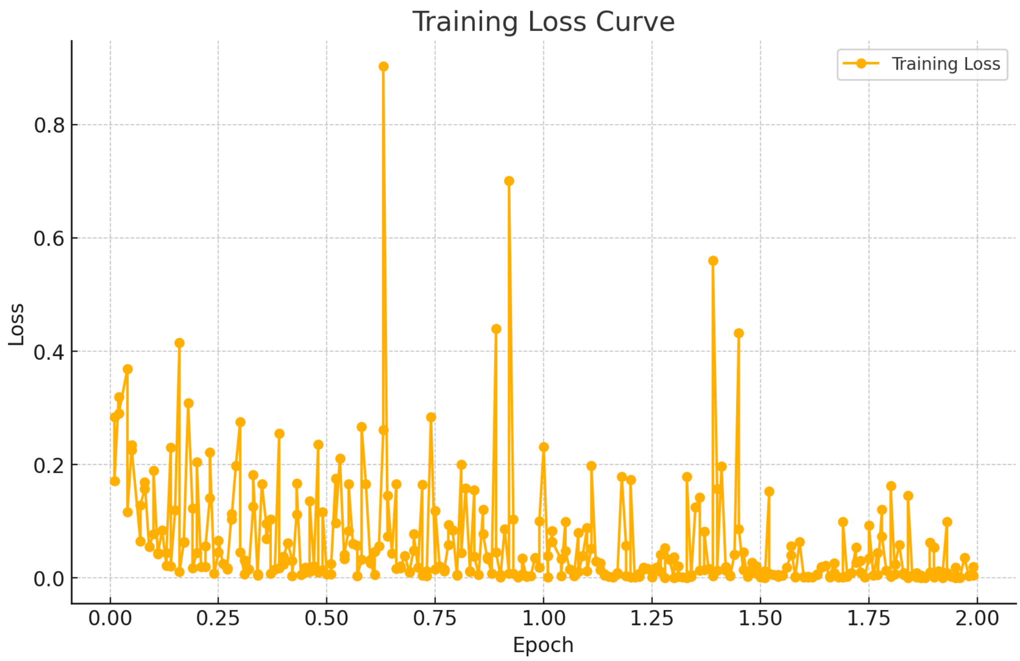
6.1 Baseline Model

The baseline model employed is InternVL-8b, a multimodal model capable of processing textual descriptions and visual inputs. It serves as a foundation for extracting and comparing product features. The **reason** for selection is that the InternVL-8b multimodal architecture enables seamless integration of image and textual information, making it well-suited for the product matching task. For the baseline model version decision, we test the different InternVL models (InternVL-2b, InternVL-4b, InternVL-8b). We test the model with 10 product information and 10 user queries to see the model's extracting ability. From the "Accuracy Comparison of Models" figure, we can see that VL-8b with the best performance. Thus, the VL8b will be the baseline model.



6.2 Fine - Tuning Model:

The model (InternVL-8b) was fine-tuned using domain-specific product data and user query to adapt the embeddings to improve the accuracy in feature extraction and output format consistency.



7 Quantitative Result

To evaluate the performance of the Product Matching System, the following metrics were used:

Product Feature Extraction Accuracy: A sample of 50 products was manually labelled with attributes like colour, dimensions, and price. The model's JSON outputs were compared to these labels, achieving 78% accuracy. Testing ensured dataset updates did not introduce errors, as incorrect extractions could lead to mismatched recommendations.

User Query Extraction Accuracy: A sample of 50 user queries was manually labelled with attributes like colour, dimensions, and price, ignoring unspecified features. The model's JSON outputs achieved 84% accuracy when compared to these labels. Testing ensured accurate extraction to align with user intent, as misinterpretations could harm product recommendations and user satisfaction.
Product Matching Accuracy: A sample of 30 user queries, each designed for a specific product and labelled with its ID, was tested. Queries included dimensions, colour, and price, and the matching pipeline returned the top 5 products. If the labelled product appeared in the results, a correct counter was incremented. The model achieved 80% accuracy, demonstrating the system's end-to-end effectiveness in recommending relevant products.

	Accuracy
Product Feature Extract	78% (39/50)
User Query Feature Extract	84% (42/50)
Pipeline Matching	**80% (24/30)**

Table 7.1 Fine-tuned Model Testing

Additionally, we tested the Baseline model using the same test dataset. From Figure 7.1, it is evident that the results of the Baseline model are inferior to those of the fine-tuned model. For the Baseline model, its output format is inconsistent, with some outputs failing to be properly converted into JSON format. Furthermore, the model occasionally misinterprets data extracted from images, such as mismatching dimensions (e.g., swapping width and height).

Figure 7.1 Baseline Model and Fine-tuned model compare

8 Qualitative Results

8.1 Product Feature Extraction

Inputs:

- Product Description: a description of the product (example below random collect from Amazon).
- Price: Image_dir: path contains the product image and will input to the model to get the dimension information.
- Product URL: Use to record the product URL path and show it to the user.

image_dir:	/content/drive/MyDrive/ECE1786/ECE1786_project/test/1
description:	SHW Desk Home Office 40-Inch Computer Table, Black
price:	88.97
product_url:	https://example.com/product

Output (JSON): Contains product information about dimension, price and colour.

{'color': 'black', 'length': 40.0, 'width': 19.0, 'height': 28.0, 'unit': 'inch', 'price': 88.97}

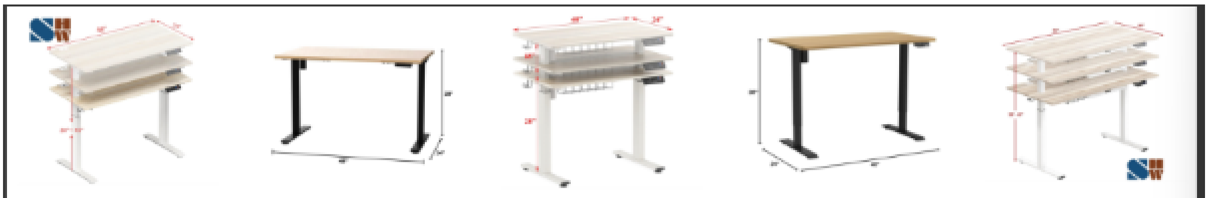
Strength: The model accurately identified the colour, dimensions, and price based on the description and clear image. This showcases the pipeline's ability to extract the product information from user input and upload it to the product database with the label.
Limitation: For products with poor-quality images or complex structure images, the model struggled to extract precede dimensions, leading to incomplete or incorrect JSON outputs. For instance, an image with poor lighting might misidentify colour (e.g., outputting "dark grey" instead of "black").

8.2 Product Matching

Inputs: The query contains target product information like dimensions, price, and colour.

user_query:	A maple table with a length of 48 inches, width of 24 inches, and height of 28 inches, priced at $249.87.

Outputs: top 5 recommendations.



	img_dir	price	link \	final_score
5	6	249.87	https://www.amazon.ca/SHW-55-Inch-Electric-Adj...	1.000000
41	42	359.99	https://www.amazon.ca/dp/B0BYY6H6CG/ref=sspa_d...	0.993581
6	7	199.87	https://www.amazon.ca/SHW-55-Inch-Electric-Adj...	0.985263
40	41	329.99	https://www.amazon.ca/dp/B08L5YQYXR/ref=sspa_d...	0.983507
4	5	299.87	https://www.amazon.ca/SHW-55-Inch-Electric-Adj...	0.980829

Figure 8.2.1 Good Product Matching Example

Strength: The model successfully extracts key information from the user's query, and translates these into a structured JSON format. This structured representation ensures the matching process is precise and aligns closely with the user's intent.
Limitation: The matching algorithm occasionally overweights certain attributes. For instance, it recommended the second product despite its price exceeding the user’s limit, which may reduce user satisfaction [The example below shows the bad case of the recommendation price over the input budget.].

user_query: maple, price about 200

```
    img_dir   price                                                link  \   final_score
6         7  199.87  https://www.amazon.ca/SHW-55-Inch-Electric-Adj...        0.699992
5         6  249.87  https://www.amazon.ca/SHW-55-Inch-Electric-Adj...        0.697093
4         5  299.87  https://www.amazon.ca/SHW-55-Inch-Electric-Adj...        0.694178
40       41  329.99  https://www.amazon.ca/dp/B08L5YQVXR/ref=sspa_d...        0.692423
41       42  359.99  https://www.amazon.ca/dp/B0BVV6H6CG/ref=sspa_d...        0.690674
```

Figure 8.2.2 Bad Product Matching Example

8.3 Comparison

Compared to the Baseline model, the fine-tuned model produces consistent outputs, reliably returning the expected JSON format. However, the Baseline model often generates outputs with irregular and unexpected formats, requiring additional processing when handling its output with code. Moreover, the information provided by the Baseline model is often inaccurate. Thus, fine-tuned InternVL-8b models have better performance than baseline models.

```
Testing 20/30
User Query: I need a small black table for my living room, under $100.
Label: {"color": "Black", "price": 100}
Result: {'color': 'Black', 'price': 100}
Matches: color=True, length=False, width=False, height=True, price=True

Testing 21/30
User Query: Looking for a wooden desk, something dark, that's not too big, around $150.
Label: {"color": "Dark", "price": 150}
Result: {'color': 'Dark Wooden', 'price': 150}
Matches: color=False, length=False, width=False, height=True, price=True
```

Figure 8.0.1. User Query Feature Extraction Result From Baseline Model

```
Testing 20/30
User Query: I need a small black table for my living room, under $100.
Label: {"color": "Black", "price": 100}
Result: {'color': 'Black', 'height': 'Small', 'price': 100}
Matches: color=True, length=False, width=False, height=True, price=True

Testing 21/30
User Query: Looking for a wooden desk, something dark, that's not too big, around $150.
Label: {"color": "Dark", "price": 150}
Error decoding JSON: Expecting value: line 1 column 1 (char 0)
Raw response: color: Dark
height: Not too big
price: $150
Result: None
Result is not a valid Python dict. Skipping comparison.
```

9 Discussion and Learning

Our final choice, the InternVL2-8b model, performed well, achieving 84% accuracy in extracting features from user prompts and 80% accuracy in recognizing features from product descriptions and images. While these results are promising, there's room for improvement. For example, the model sometimes struggles with subtle details, like distinguishing between similar colours or handling products with multiple features.

One key takeaway is how much the dataset quality impacts performance. A more diverse and larger dataset would likely help the model handle real-world variability better. If we were to start over, we'd focus on creating a balanced dataset from the start and using feedback to refine the model further.

Thus, the model shows strong potential but highlights the need for better data and refinement to handle more complex scenarios effectively.

10 Individual Contributions

Jiaguan Tang

- Model Fine-tuning and Hyperparameter Optimization: Fine-tuned the model and adjusted hyperparameters to optimize performance.
- Product Matching Code: Developed the algorithm for matching products with user queries.
- Prompt Design: Designed prompts to improve model matching accuracy.

Yue Lan

- Dataset creation with multimodal product features and user queries
- Processed and structured product data into JSON format
- Developed preprocessing pipelines for feature extraction using regex
- Conducted research and compiled related work for the project

Chen Wang

- "Final matching algorithm" - design the final algorithm to match output features with the ground truth labels.
- "Training and test Dataset Create" - Collect and manually label 50 product data, generate and manually label 50 user queries.
- Design Proposal and Progress Report (everything except the data section)

Bocheng Zhang

- "Model Implement" - Implement the InternVL-8b model for both product & user query feature extraction.
- "Model Comparison" - Designing code to compare the "VL-2b", "VL-4b", and "VL-8b" model's performance.
- "Training & Testing Dataset Create" - Collect 50 product data. Generate 50 user queries. Design 30 matching test datasets.
- "Documentation" - Test the model and build the result table/ figure. Write the Result Section. Draw the pipeline graph.

Reference

1. Lin, R., He, X., Feng, J., Zalmout, N., Liang, Y., Xiong, L., Dong, X. L. (2021). PAM: Understanding Product Images in Cross-Product Category Attribute Extraction. arXiv preprint arXiv:2106.04630. Retrieved from https://arxiv.org/abs/2106.04630
2. Liu, C., Hou, P., Zeng, A., Yu, H. (2023). Transformer-empowered Multi-modal Item Embedding for Enhanced Image Search in E-Commerce. arXiv preprint arXiv:2311.17954. Retrieved from https://arxiv.org/abs/2311.17954